How this book is organized: A road map

Although this book is not divided into parts, the chapters follow a clear linear progression, beginning with an introduction to Elasticsearch from a feature and architecture perspective:

- Chapter 1 embarks on a journey through the world of search, retracing the steps from rudimentary database-backed systems to the sophisticated search engines that are the norm today. We shine a spotlight on Elasticsearch, a powerful, versatile modern search engine that has redefined the capabilities of search functions, bringing to the forefront its distinct features, real-world applications, and widespread adoption.

  We also look ahead to the transformative potential of general artificial intelligence tools. We examine the exciting possibilities of technologies like ChatGPT, exploring how they could reshape the search space and redefine our interactions with information in the future.
- Chapter 2 dives headfirst into Elasticsearch, providing hands-on experience with indexing and retrieving documents using the document APIs. We also execute search queries using the search APIs. The chapter takes you through essential search criteria, from pattern matching to phrase searches, spelling corrections, range results, multi-field searches, and more. A glimpse at advanced queries further enriches the learning experience. The chapter concludes with a look at data sorting, result pagination, highlighting, and other impressive functionalities that elevate the user's search capabilities.
- Chapter 3 demystifies the architecture of Elasticsearch, guiding you through its foundational components and the intricate processes that enable searching and indexing. This exploration covers fundamental concepts that power the search engine, including the inverted index, relevancy, and text analysis. We also explore clustering and the distributed nature of the Elasticsearch server.
- Chapter 4 explores mapping schemas, data types, and mapping APIs, providing a detailed overview of data handling in Elasticsearch. The chapter looks at how mapping schemas enhance search accuracy and efficiency, thoroughly examining dynamic and explicit mapping. The exploration extends to core data types, including `text`, `keyword`, `date`, and `integer`. The chapter concludes with advanced data types such as `geo_point`, `geo_shape`, `object`, `join`, `flattened`, and more.
- Chapter 5 offers a comprehensive discussion of single- and multi-document APIs and their associated operations. This chapter provides a practical understanding of indexing, retrieving, updating, and deleting documents using these APIs. It also explores the reindexing feature.
- Chapter 6 zooms in on indexing operations using the indexing APIs. It also guides you through the foundational configurations of an index, including settings, mapping, and aliases. This exploration provides an understanding of